



Indian Institute of Information Technology (IIIT) Surat
Department of Computer Science & Engineering
B.Tech-II, Semester - III
Mid Semester Examination, September-2025
Subject: Database Management Systems (CS 302)

Date: 23rd September, 2025

Timing: 03:30 P.M. to 05:00 P.M.

Max Marks: 30

Instructions:

1. Make suitable assumptions whenever necessary with proper justifications.
2. Figures to the right indicate maximum marks assigned for that question.
3. **Attempt all questions.**

- Q.1. Explain the different levels of data abstraction in a DBMS and provide examples for each level using the university database scenario. [3]
- Q.2. List four significant differences between a file-processing system and a DBMS with example. [4]
- Q.3. Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period of time, and has an associated due date, and the date when the payment was received. Draw the E-R diagram for the given car insurance company scenario, showing all entities, attributes, and relationships with appropriate cardinalities. Clearly indicate primary keys, candidate keys, and foreign keys, and represent the possible relationships among the entities if any. [6]
- Q.4. Consider the relational schema for the *employee* where the primary keys are underlined as given below. Give an expression in the relational algebra to express each of the following queries: [5]

employee(ID, person_name, street, city)
works(ID, company_name, salary)
company(company_name, city)
manages(ID, manager_id)

- Find the names of all employees whose salary is greater than 100,000.
 - Find the names of all employees who live in "Mumbai" and whose salary is greater than 150,000.
 - Find the ID and name of each employee who does not work for "BigBank".
 - Find names of employees who work for at least one company located in "Hyderabad".
 - Find IDs of employees who work for every company.
- Q.5. Consider the above given relational employee schema. Give an expression in the Tuple Relational Calculus (TRC) to express **TWO** of the following queries: [4]
- For each company located in "Delhi", find the names of employees who belong to that company.
 - Names of employees who do not work for any company.

- Names of employees who live in the same city as at least one employee with *salary* > 200000.

Q.6. Consider the above given relational employee schema. Write a SQL query for **FOUR** the following queries:

[8]

- Find the names and cities of residence of all employees who work for "First Bank Corporation".
- Find the names, street, and cities of residence of all employees who work for "First Bank Corporation" and earn more than 10,000.
- Find all employees in the database who earn more than each employee of "Small Bank Corporation".
- Assume that the companies may be located in several cities. Find all companies located in every city in which "Small Bank Corporation" is located.
- Find those companies whose employees earn a higher salary, on average, than the average salary at "First Bank Corporation".